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Season 4

THINK. BUILD. CREATE. WIN

CASE STUDY 1

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Self-Healing Machines: Predictive Maintenance for a Connected Fleet

Background & Problem Statement

A fleet of connected machines - trucks, buses, or industrial compressors - continuously streams sensor data such as vibration, temperature, motor current, pressure and RPM. Today, failures are mostly caught after a breakdown or through fixed service intervals, leading to unplanned downtime and higher maintenance cost.

The team must build a closed-loop system that ingests this sensor data, detects early signs of degradation, predicts how much useful life a machine has left, and then automatically raises a maintenance work order - with a human stepping in only for exceptions.

What the Team Must Demonstrate

Layer	What to build
Data Engineering	Read the sensor data (batch or streaming), clean it - handle missing values and basic noise - and create simple rolling-window features (e.g., averages, trends) for the model.
Machine Learning	Detect abnormal behavior and predict the Remaining Useful Life (RUL) - how many cycles a machine has left before failure.
MLOps	Track experiments and model versions and show a simple retraining flow when new data arrives, with a basic monitoring view.
Agents (non-LLM)	A rule-based Triage Agent that reads the model output, decides the severity, and automatically creates a maintenance work order. No LLM required.

Ground Rules

- Failures are less common than normal reading, so the model should handle mild imbalance.
- The pipeline should run end-to-end without manual steps once data is provided.
- Keep it lightweight - no LLM. The agent should use simple rules or thresholds.
- Use open-source tools; document the design clearly so it can be explained in a review.

Dataset

Use the **NASA C-MAPSS Turbofan Engine Degradation** dataset - a well-documented benchmark of medium complexity. It contains run-to-failure sensor readings from a fleet of engines (21 sensors + 3 operating settings), with true Remaining Useful Life values provided for evaluation.

The dataset comes in four subsets (**FD001–FD004**) of increasing difficulty. You may choose to work with just one subset (**FD001** is the simplest, and a good starting point) or take on all four for a more challenging solution.

Official (NASA) | data.nasa.gov/dataset/cmapss-jet-engine-simulated-data

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Round-wise Deliverables

Round 1 (Idea Submission) – Design deck

- Architecture diagram: data → features → model → agent → work order
- Approach for cleaning data and building features
- Choice of anomaly-detection and RUL models, with reasoning
- Logic the Triage Agent will use

Round 2 (Detailed Submission) - Proof of Concept

- Working pipeline on the chosen C-MAPSS subset(s)
- Trained model with RUL predictions
- A simple monitoring dashboard
- Agent auto-generating a work order + a demo retraining run

Suggested Tools

These are optional starting points - you are free to use alternatives. The choice of models and algorithms is left entirely to the team.

Layer	Suggested tools / libraries
Data Engineering	Python, Pandas, NumPy; optionally PySpark for larger-scale processing
Machine Learning	scikit-learn; optionally TensorFlow or PyTorch
MLOps	MLflow for experiment and model tracking; Streamlit or Dash for a simple dashboard
Agent	Plain Python for rule-based logic - no LLM or agent framework required

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Evaluation Criteria

Round 1 - Evaluation of Design

Round 1 submission are design and Architecture slides deck. The evaluation at this phase focuses on the **clarity, completeness and soundness of the proposed design.**

Criterion	Weight	Assessed on
Problem understanding	15%	Demonstrates a clear and accurate grasp of the problem and its objectives.
Architecture clarity	20%	Presents a coherent, well-organized design with clearly defined components and how they connect.
Data engineering approach	15%	Proposes a sound, well-reasoned approach to preparing and transforming the input data.
ML approach & reasoning	20%	Selects suitable methods and justifies them with clear reasoning; shows awareness of practical challenges.
MLOps & agent logic	15%	Describes a practical operational plan and a clear, logical decision flow for the agent.
Dataset coverage	10%	Design accounts for the varying complexity of the available data and scales its approach accordingly.
Communication quality	5%	Well-structured, unambiguous and internally consistent.

Round 2 - Evaluation of Proof of Concept

Round 2 is a working build, and the review focus shifts from plan to execution.

Criterion	Weight	Assessed on
Working pipeline	20%	Runs end-to-end and produces results with minimal manual intervention.
Data engineering quality	15%	Data handling is clean, robust and reproducible.
Model performance	15%	Produces credible results with appropriate, sensibly interpreted metrics.
MLOps in action	10%	Operational elements are functional and demonstrable.
Agent behavior	10%	The agent acts correctly and reliably on the model output.
Dataset coverage	10%	Depth and breadth of data tackled relative to the available complexity.
Demo & explanation	20%	Team articulates its choices, trade-offs and potential improvements convincingly.